

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

Paper 1

Monday 13 May 2024

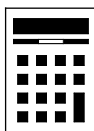
07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



J U N 2 4 9 2 0 2 1 0 1

Answer **all** questions in the spaces provided.

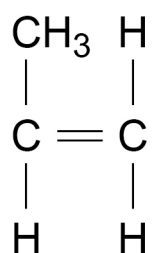
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0 1

Propene is an alkene.

Figure 1 represents a molecule of propene (C_3H_6).

Figure 1



0 1 . 1

What is the general formula for alkenes?

[1 mark]

Tick (✓) **one** box.

C_nH_n

☐

C_nH_{n+2}

☐

C_nH_{2n}

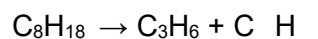
☐

C_nH_{2n+2}

☐


Octane (C_8H_{18}) can be cracked to produce propene (C_3H_6) and **one** other product.

The equation for the reaction is:



0 1 . 2 What is the formula of the other product?

[1 mark]

Tick (✓) **one** box.

C_2H_6

☐

C_3H_8

☐

C_4H_{10}

☐

C_5H_{12}

☐

0 1 . 3 What type of reaction is cracking?

[1 mark]

Tick (✓) **one** box.

Combustion

☐

Decomposition

☐

Neutralisation

☐

Oxidation

☐

Question 1 continues on the next page

Turn over ►



Many molecules of propene join together to produce the polymer poly(propene).

0 1 . 4

What name is given to molecules that join together to produce a polymer?

[1 mark]

Tick (✓) **one** box.

Alkanes

☐

Bases

☐

Monomers

☐

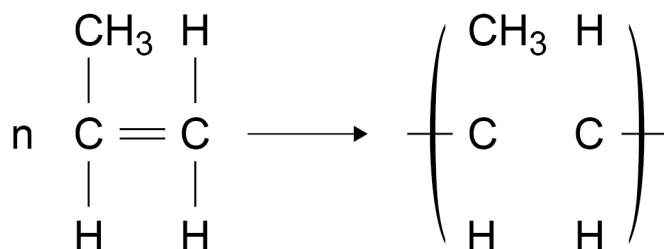
Precipitates

☐

0 1 . 5

Complete the equation to show the formation of poly(propene) from propene.

[2 marks]



0 1 . 6

Give **one** property of poly(propene) that leads to problems with waste disposal.

[1 mark]

0 1 . 7

Give **one** environmental problem caused by the disposal of poly(propene).

[1 mark]

8

Turn over for the next question

Turn over ►



0 2

This question is about ammonia.

0 2**1**

What is the test for ammonia gas?

[1 mark]Tick (✓) **one** box.

Pops when a lighted splint is placed near

☐

Relights a glowing splint

☐

Turns damp red litmus paper blue

☐

Turns limewater cloudy white

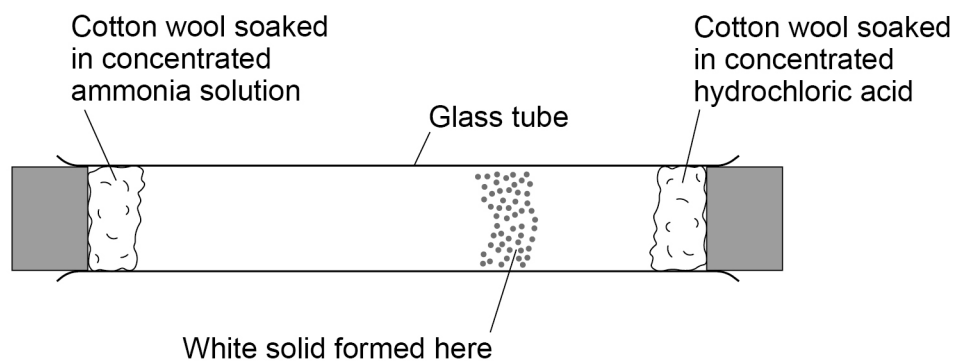
☐

A teacher demonstrated the movement of different gas molecules.

The teacher used:

- concentrated ammonia solution that gives off ammonia gas
- concentrated hydrochloric acid that gives off hydrogen chloride gas.

Ammonia gas and hydrogen chloride gas react to produce a white solid.

Figure 2 shows the apparatus.**Figure 2**

0 2 . 2

Ammonia molecules and hydrogen chloride molecules move along the tube in **Figure 2**.

What is the name of the movement of molecules along the tube?

[1 mark]

Tick (✓) **one** box.

Diffusion

☐

Fermentation

☐

Precipitation

☐

0 2 . 3

What is the name of the white solid formed in **Figure 2**?

[1 mark]

Tick (✓) **one** box.

Ammonium chloride

☐

Ammonium sulfate

☐

Sodium chloride

☐

Sodium sulfate

☐

0 2 . 4

Ammonia has a smaller relative formula mass than hydrogen chloride.

Why does the white solid form at the position shown in **Figure 2**?

[1 mark]

Question 2 continues on the next page

Turn over ►



0 2 . 5 Ammonia dissolves in water to produce an alkaline solution.

Which ion makes solutions alkaline?

[1 mark]

Tick (✓) **one** box.

H⁺

☐

O²⁻

☐

OH⁻

☐

0 2 . 6 52.0 g is the maximum mass of ammonia that dissolves in 100 g of water.

Calculate the minimum mass of water needed to dissolve 10.4 g of ammonia.

[2 marks]

Minimum mass of water = _____ g



0 2 . 7

A solution of ammonia has a concentration of 0.06 mol/dm^3 .

Calculate the mass of ammonia (NH_3) in 8.0 dm^3 of this solution.

Relative atomic masses (A_r): N = 14 H = 1

[4 marks]

Mass of ammonia = _____ g

11

Turn over for the next question

Turn over ►



0 3

This question is about structure and bonding.

0 3 . 1

Which element is diamond a form of?

[1 mark]Tick (✓) **one** box.

Calcium

☐

Carbon

☐

Oxygen

☐

Nitrogen

☐**0 3 . 2**Sodium (Na) reacts with chlorine (Cl₂) to produce sodium chloride (NaCl).

What is the balanced symbol equation for the reaction?

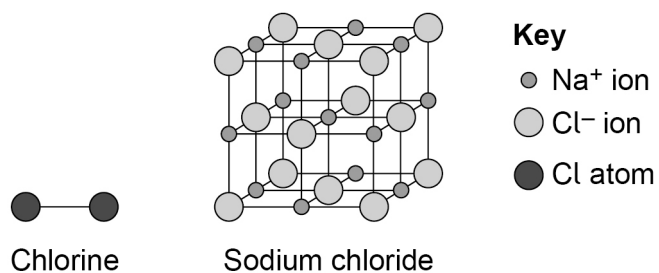
[1 mark]Tick (✓) **one** box.Na + Cl₂ → NaCl☐2Na + Cl₂ → NaCl☐2Na + Cl₂ → 2NaCl☐

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0	3	.	3
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Figure 3 represents the structure of chlorine and of sodium chloride.

Figure 3



Compare the structure and bonding of chlorine with the structure and bonding of sodium chloride.

[6 marks]

[illegible]

Turn over for the next question

8

Turn over ►



0 4

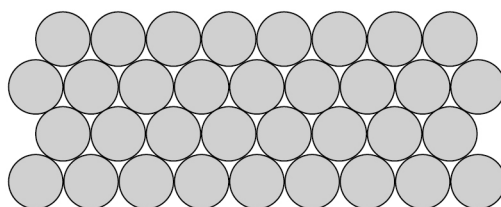
This question is about metals.

0 4 . 1

Complete the sentence.

[1 mark]

Metals conduct heat and electricity because metals contain delocalised

0 4 . 2**Figure 4** represents the structure of a metal.**Figure 4**

Describe why metals can be bent or hammered into shape.

[2 marks]

Lead and tin are metals.

Solder is a mixture of lead and tin.

0 4 . 3

What name is given to a mixture of metals?

[1 mark]



Table 1 shows how changing the percentage (%) of tin in solder affects the properties of solder.

Table 1

Percentage (%) of tin in solder	Strength in arbitrary units	Melting point in °C
0	12	328
10	18	297
20	31	257
30	34	217
40	37	187
50	41	147
60	38	153
70	35	167

0 4 . 4

Describe how the strength of solder changes as the percentage of tin in solder changes.

Use data from **Table 1**.

[3 marks]

0 4 . 5

What is the melting point of lead?

Use **Table 1**.

[1 mark]

8

Turn over ►



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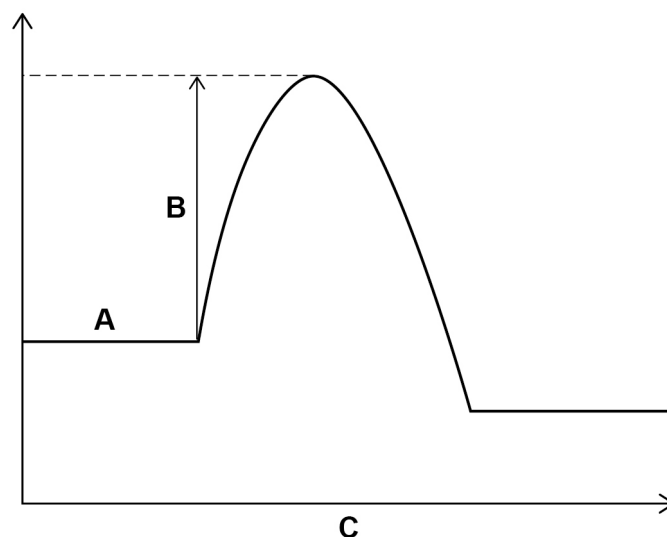
0 5

The reaction between sulfuric acid and sodium hydroxide solution is exothermic.

0 5 . 1

Figure 5 shows an energy level diagram for this reaction.

Figure 5



What do labels **A**, **B** and **C** represent on **Figure 5**?

Choose answers from the box.

[3 marks]

activation energy	energy	overall energy change
products	progress of reaction	reactants

A _____

B _____

C _____

Question 5 continues on the next page

Turn over ►



A student investigated the temperature change when sulfuric acid is added to sodium hydroxide solution.

This is the method used.

- 1 Measure 50 cm³ of sodium hydroxide solution using a measuring cylinder.
- 2 Add the sodium hydroxide solution to a plastic cup.
- 3 Measure the temperature of the sodium hydroxide solution.
- 4 Add 50 cm³ of sulfuric acid to the sodium hydroxide solution.
- 5 Stir the solution.
- 6 Measure the temperature of the mixture every minute for 10 minutes.

Figure 6 shows the apparatus.

Figure 6

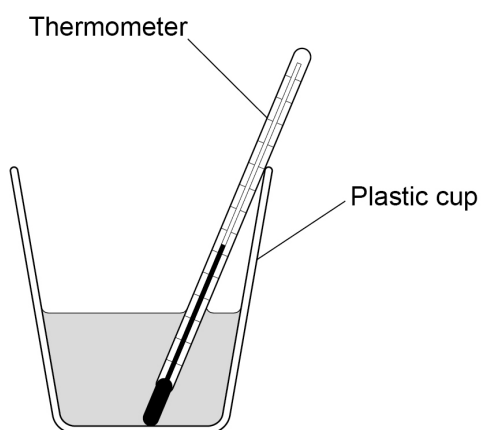
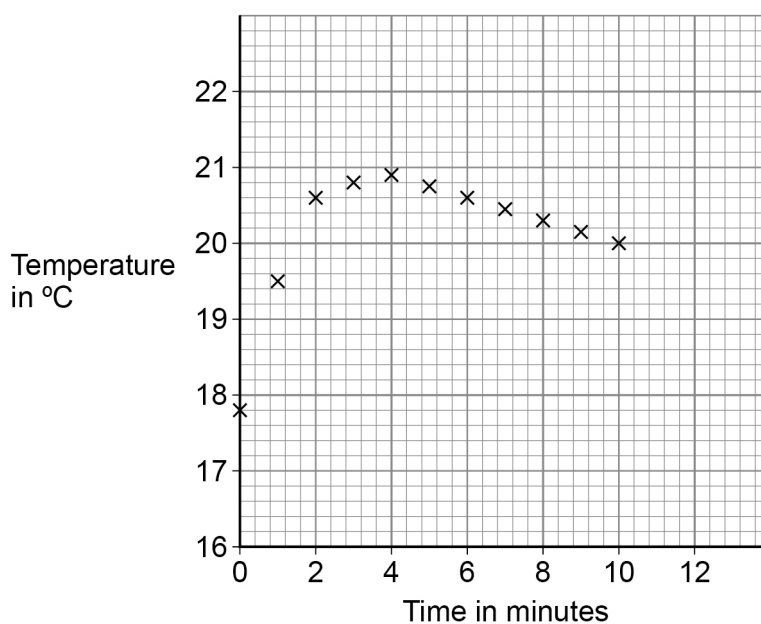


Figure 7 shows the results.

Figure 7



0 5 . 2 Complete **Figure 7**.

You should:

- draw a line of best fit through the points from **4** to **10** minutes
- extend the line of best fit back to the y-axis.

[2 marks]

0 5 . 3 The maximum temperature reached is found at the point where the line of best fit meets the y-axis.

Determine the temperature change for the reaction.

Use **Figure 7**.

[2 marks]

Temperature change = _____ °C

0 5 . 4 Explain the trends shown on **Figure 7**.

Refer to energy transfers in your answer.

[4 marks]

Question 5 continues on the next page

Turn over ►



0 5 . 5 Suggest **one** improvement to the investigation.

Give **one** reason why this would improve the investigation.

[2 marks]

Improvement _____

Reason _____

0 5 . 6 In the initial investigation, the student used 50 cm³ of 1.0 mol/dm³ sulfuric acid.

The student repeated the investigation using 25 cm³ of 2.0 mol/dm³ sulfuric acid.

The number of moles of sulfuric acid was the same in each investigation.

The maximum temperature recorded was higher in the repeated investigation.

Explain why the maximum temperature recorded was higher.

[2 marks]

15



0 6

This question is about the extraction of copper.

0 6 . 1

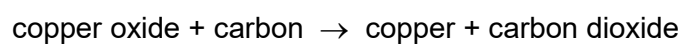
Copper is a useful metal.

Give **one** use of copper.

[1 mark]

Copper can be extracted from copper oxide using carbon.

The equation for the reaction is:



This is a redox reaction.

0 6 . 2

Which **two** processes happen in a redox reaction?

[1 mark]

 and

0 6 . 3

Explain why carbon can be used to extract copper from copper oxide.

[2 marks]

Question 6 continues on the next page

Turn over ►

0	6	.	4
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Copper oxide contains 79.9% by mass of copper.

A sample of copper oxide contains 11.8 g of copper.

Calculate the mass of the sample of copper oxide.

Give your answer to 3 significant figures.

[4 marks]

Mass of sample of copper oxide (3 significant figures) = _____ g



Copper can be extracted from low-grade ores using bacteria or plants.

0 6 . 5

Give **one** reason why scientists are researching new ways to extract copper.

[1 mark]

0 6 . 6

Name the process that uses bacteria to extract metals.

[1 mark]

0 6 . 7

Give **one** advantage and **one** disadvantage of using bacteria to extract copper compared with traditional methods.

[2 marks]

Advantage _____

Disadvantage _____

0 6 . 8

Phytomining uses plants to extract copper compounds from low-grade ores.

Describe how phytomining is used to extract copper compounds.

[3 marks]



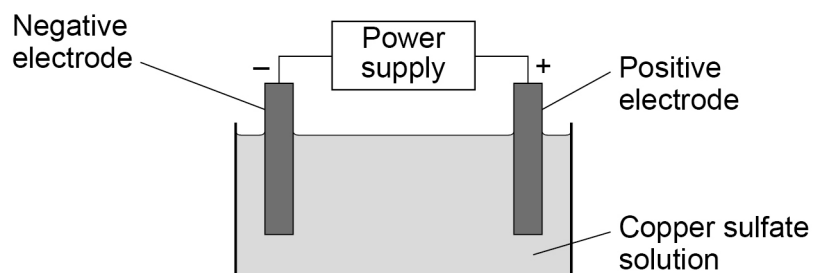
0 7

This question is about electrolysis.

Figure 8 shows the apparatus used for the electrolysis of copper sulfate solution.

Copper sulfate contains copper ions and sulfate ions.

Figure 8



The negative electrode becomes coated with copper.

0 7 . 1

What is the name of the substance broken down in electrolysis?

[1 mark]

Tick (✓) **one** box.

Electrode

☐

Electrolyte

☐

Electron

☐

A student measured how the mass of the negative electrode changes with time.

This is the method used.

- 1 Measure the mass of the negative electrode.
- 2 Set up the apparatus shown in **Figure 8**.
- 3 Turn on the power supply for 10 minutes.
- 4 Remove the negative electrode and place briefly in a beaker of liquid propanone.
- 5 Dry the negative electrode by letting the propanone evaporate.
- 6 Measure the mass of the negative electrode.

0 7 . 2

Propanone is a flammable liquid.

Give **one** safety precaution you should take when using propanone.

[1 mark]

0 7 . 3

Propanone has a low boiling point.

Suggest why propanone is used to dry the electrode.

[1 mark]

0 7 . 4

Explain why the negative electrode must be dried before recording the mass.

[2 marks]

Question 7 continues on the next page

Turn over ►



0 7 . 5

Copper is deposited on the negative electrode at a rate of 0.005 g/s.

Calculate the time taken for the mass of the negative electrode to increase by 4.65 g.

[2 marks]

Time = _____ s

0 7 . 6

Copper sulfate solution is blue.

How does the appearance of the copper sulfate solution change during the experiment?

Give **one** reason for your answer.

[2 marks]

Change in appearance _____

Reason _____



0	7	.	7
---	---	---	---

The student investigated the electrolysis of zinc sulfate solution.

Bubbles of gas were produced at the negative electrode.

Explain why the gas was produced at the negative electrode.

You should identify the gas.

[3 marks]

12

Turn over for the next question

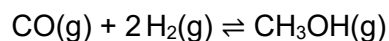
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0	8
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Methanol (CH₃OH) is produced by reacting carbon monoxide with hydrogen.

The equation for the reaction is:



The forward reaction is exothermic.

0	8	.	1
---	---	---	---

Name the type of energy change in the backward reaction.

[1 mark]

0	8	.	2
---	---	---	---

Explain the effect of decreasing the temperature on the yield of methanol at equilibrium.

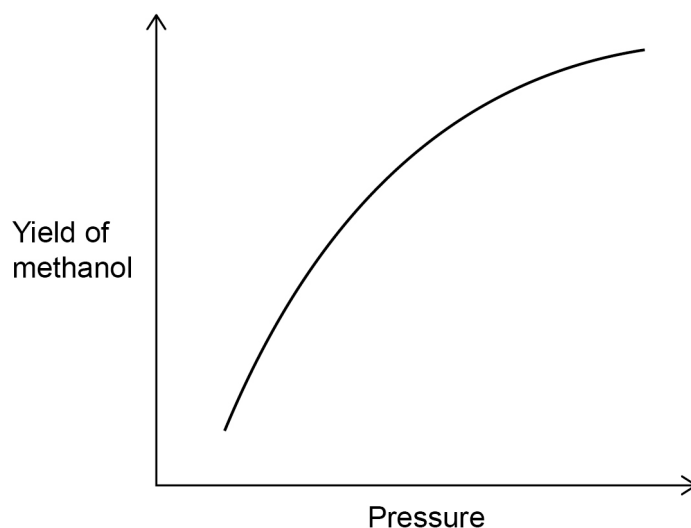
[2 marks]



0 8 . 3

Figure 9 shows how the yield of methanol changes when the pressure of the equilibrium mixture is increased.

Figure 9



Explain the change in the yield of methanol when the pressure is increased.

Use **Figure 9** and the equation for the reaction.

[3 marks]

Question 8 continues on the next page

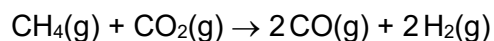
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0	8	.	4
---	---	---	---

Carbon monoxide (CO) can be produced from methane (CH₄).

The equation for the reaction is:



1440 dm³ of methane is completely reacted with carbon dioxide.

Calculate the maximum mass of carbon monoxide produced.

Give your answer in kilograms.

The volume of one mole of a gas is 24.0 dm³.

Relative formula mass (M_r): CO = 28

[5 marks]

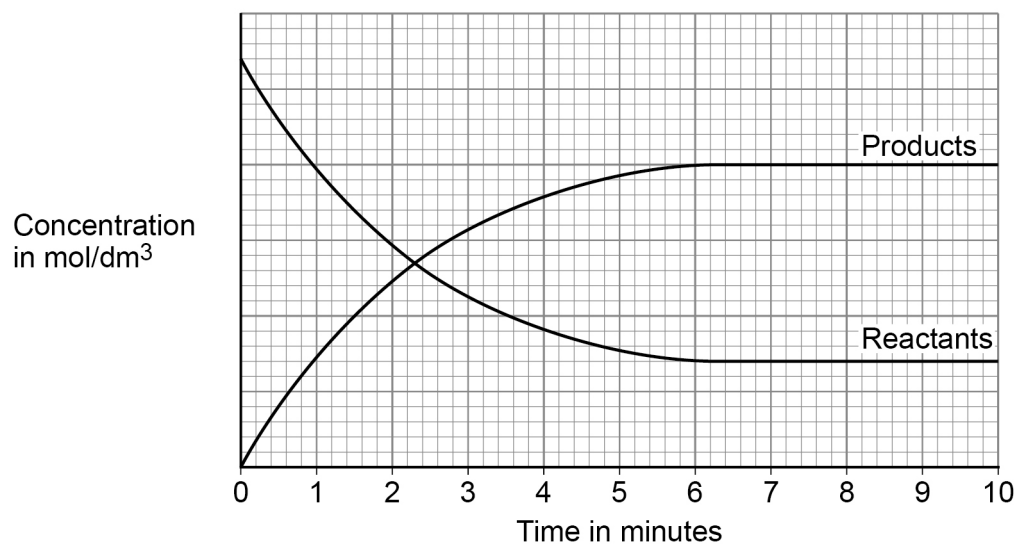
Mass = _____ kg



0 8 . 5

Figure 10 shows how the relative concentrations of the reactants and the products change with time in an equilibrium reaction.

Figure 10



Determine the time when equilibrium is first reached.

Give **one** reason for your answer.

[2 marks]

Time _____

Reason _____

13

END OF QUESTIONS



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